

Construction of Safety Inventory Model for Specialized Coal Terminals under Uncertain Demand

1st Jinpeng Liu
SDIC Caofeidian Port Co., Ltd.
Tangshan, China
liujinpeng@scfdport.com

2nd Xuliang Tang
Shanghai Institute of Technology
Shanghai, China
fayen49@qq.com

3rd Yong Huang
Hangzhou Branch of Shanghai Jiudao
Information Technology Co., Ltd.
Hangzhou, China
huangyong@jiudaotech.com

Abstract—Safe stock is an important guarantee for the normal production of the port. Port materials belong to equipment operation supporting materials, with the characteristics of uncertain demand. Based on the actual business situation of Caofeidian professional coal terminal, this paper conducts targeted research on different types of materials. Based on the general safety stock model, the safety stock model is improved and optimized, and the safety stock management model applicable to Caofeidian professional coal terminal is established.

Keywords—uncertain demand, Safety stock, professional coal terminal

I. INTRODUCTION

The inventory cost of spare parts and materials accounts for a large proportion of the total cost of Caofeidian professional coal terminal [1]. Caofeidian professional coal terminal has a wide variety of spare parts and materials, about 10000 kinds. This paper takes the 10-year historical procurement data and consumption data of Caofeidian professional coal terminal in the asset management system as samples, classifies the spare parts and materials, and establishes different classification safety inventory prediction models [2]. It has practical significance for inventory cost control of Caofeidian professional coal terminal [3].

With the expansion of Caofeidian specialized coal terminal market in recent years, the production volume has increased continuously, and the demand for spare parts and materials has increased rapidly [4]. The inventory cost in 2020 increased by 32 million yuan or 57% compared with that in 2018 [5]. In the actual business of Caofeidian Professional Coal Terminal, the procurement personnel checked and reduced the procurement demand based on their personal experience and the inventory quantity, but the actual control effect was poor due to the variety of materials and the defects of personal experience. In the port industry, the future demand for spare parts and materials as production support materials is uncertain. If the inventory is kept at a low level, there will be a risk of equipment downtime and production losses. How to maintain an appropriate inventory, reduce costs, and ensure that the inventory can guarantee production to the level expected by the enterprise is an urgent problem to be solved.

For safety inventory, many companies have a more extensive method of setting it, and they mostly set a certain period of usage as safety inventory based on experience. For example, for product A with a production cycle of 4 weeks, the estimated sales quantity for 4 weeks is set as safety stock.

There are two issues here. Firstly, in practical operation, the safety stock level for these four weeks is often determined by experience and a pat on the head. It is undeniable that these experiential values often embody a lot of organizational wisdom, which is simple and easy to implement. However, since it is based on experience, it is destined to have its own problems. In many enterprises, with a wide range of product specifications, it is necessary to set a certain amount of safety stock [6]. If such a large amount of data is set based on experience, it is bound to cause surplus, shortage, and coexistence of shortage and surplus, which is a typical planning problem. If the knowledge of mathematical statistics can be used to quantify the uncertainty of the demand and supply sides, then the demand can be predicted more accurately, and a more reasonable setting of safety inventory can be known. Secondly, the safety inventory set in this way often sacrifices the inventory amount to achieve 100% service level, without considering the hierarchical requirements of service level in a more detailed manner [7]. One indicator for quantifying service levels is the availability rate of products. If the availability rate is too low, it will affect revenue, while if it is too high, it will require bearing inventory cost risks. Therefore, in order to scientifically set the quantity of safety stock, service level indicators must also be considered [8].

The prediction of safety stock of spare parts and materials is of great significance to the construction of intelligent operation and maintenance of equipment [9]. The prediction of spare parts and material safety inventory can help the equipment intelligent operation and maintenance system prepare the required spare parts in advance, avoid equipment downtime due to the lack of spare parts and materials, and thus improve the equipment reliability [10]. The safety stock prediction of spare parts and materials can enable the equipment intelligent operation and maintenance system to conduct intelligent procurement according to the demand of spare parts and materials, avoid the occupation and waste of funds due to the backlog of spare parts and materials, and thus reduce the equipment maintenance cost. The safety stock prediction of spare parts and materials can enable the equipment intelligent operation and maintenance system to carry out intelligent distribution according to the demand of spare parts, improve the distribution efficiency and maintenance efficiency of spare parts and materials, shorten the equipment downtime, and reduce production losses.

II. GENERAL MODEL OF SAFETY STOCK

Safety Stock (SS), also known as safety stock reserve, is a buffer stock built to meet the expected consumption in a given

period. This is a kind of inventory held in addition to meeting the average demand, and its purpose is to maintain the expected supply level of goods. The determination of safety stock is based on the consumption of previous years, based on mathematical statistics, and based on the assumptions of demand and LT. According to the contents of Junyi Kobayashi's "Inventory Management" and the relevant calculation methods of safety stock theory, assuming that the average demand and order lead time of inventory goods in the supply system are random, and the average demand and order lead time obey normal distribution, the general model of safety stock is set as follows:

$$SS = z \sqrt{\sigma_d^2(\bar{L}) + \sigma_L^2(\bar{d})^2} \quad (1)$$

In the formula, $\bar{L}$ is average value of lead time. $\bar{d}$ is average daily demand. σ_d is the standard deviation of daily demand d , which is not related to the same normal distribution as the average demand is σ_L . The standard deviation of the lead time L follows the normal distribution with the lead time. Z is the number of standard deviations under a certain service level is the safety factor.

Z represents the value of the standard deviation of the average corresponding to the specific supply level. For example, the service level of 95.00% means that there is a probability of 0.05 when the demand is greater than the safety stock during the delivery period of goods. The safety factor Z corresponding to the shortage rate is 1.65. If the safety factor is set too large, the safety stock holdings will be too large. Generally, the supply level is set at 95.00% - 99.00%, and the value of Z is 1.65 - 2.33. In relevant literature, Z value is usually given directly as a known quantity.

TABLE I. SAFETY FACTOR CORRESPONDING TO COMMON SUPPLY LEVEL

Supply level (%)	Shortage rate (%)	Corresponding safety System Z
100.00	0.00	∞
99.99	0.01	3.72
99.50	0.50	2.81
99.00	1.00	2.33
98.00	2.00	2.05
97.00	3.00	1.88
95.00	5.00	1.65
90.00	10.00	1.28
85.00	15.00	1.04
84.00	16.00	1.00
80.00	20.00	0.84
75.00	25.00	0.68
60.00	40.00	0.38
50.00	50.00	0.17

III. CONSTRUCTION OF SAFETY STOCK MODEL

The general model of safety stock is only applicable to the ideal situation of inventory management, that is, stable equilibrium of consumption and supply. In the actual situation of specialty A coal terminal, the consumption of spare parts and most of the materials is affected by various factors such as production, equipment status, equipment technical

transformation plan, equipment overhaul plan, etc., and the purchase demand is affected by institutional and human factors. Therefore, the general safety stock model cannot adapt to the actual situation of the site, so it is necessary to improve the safety stock model to make it more consistent with the actual situation of the A professional coal terminal.

A. Safety stock classification

A professional coal terminal inventory mainly provides spare parts for equipment maintenance, materials for equipment maintenance, fuel for equipment operation and office and labor protection supplies for daily use. According to the characteristics of inventory economy and consumption, the inventory is divided into three categories: key spare parts, seasonal materials and general materials.

a) *Key spare parts*: Spare parts with high value, low consumption frequency, long supplier cycle, irregular consumption frequency and supply cycle.

b) *Seasonal materials*: Materials with high consumption frequency and obvious seasonal characteristics.

c) *General materials*: Office supplies, labor protection supplies and other auxiliary materials with low value, high consumption frequency, short supplier cycle, regular consumption frequency and supply cycle.

For different categories of spare parts or materials, different safety stock models will be used to predict the safety stock. Compared with the static calculation method of the general safety stock model, the classification calculation will better reflect the actual situation of the operation and management of professional A coal terminal

B. General material safety inventory model

General materials for professional coal terminal A mainly include daily office supplies, labor protection supplies and other auxiliary materials. The daily consumption of such materials has strong regularity. The daily procurement channels are e-commerce platforms such as JD and the industrial procurement platform, and the supply cycle is short and stable. Through the analysis of the historical data of general materials, if the daily statistical cycle is used, the demand history no longer conforms to the normal distribution, and finally the monthly statistical cycle is selected for calculation. The data is less discrete, which conforms to the general model theory of safety stock. At the same time, the method of monthly statistical cycle is also consistent with the business model of monthly submission of procurement plan by professional A coal terminal. Therefore, by improving the general model of safety stock, the safety stock model of general materials for professional A coal terminal can be obtained.

C. Seasonal material safety inventory model

A professional coal terminal is located in the Bohai Bay area. According to the local climate characteristics and daily material consumption, the 12 months of the year are divided into summer and winter, with June to October in summer and November to May in winter.

TABLE II. SAFETY FACTOR CORRESPONDING TO COMMON SUPPLY LEVEL

Month	Average daily maximum temperature (°C)	Average daily maximum temperature (°C)
Jan.	3	-8
Feb.	4	-8
Mar.	13	0
Apr.	22	8
May.	25	10
Jun.	31	19
Jul.	32	23
Aug.	31	22
Sept.	27	16
Oct.	20	7
Nov.	10	-2
Dec.	3	-9

Based on the historical consumption data of spare parts and materials for 10 years, spare parts and materials with seasonal characteristics are selected. In order to eliminate randomness, spare parts and materials with less than 3 historical consumption times are eliminated from the sample data. Convert the consumption date of spare parts and materials that have been consumed more than 3 times into seasons, and count the season in which each consumption leaves the warehouse. 100% of the time spent in the warehouse in summer or winter is defined as seasonal spare parts or materials.

TABLE III. SPARE PARTS: 90-DEGREE GALVANIZED THREADED ELBOW AND DN32 HISTORICAL CONSUMPTION DATA

Delivery month	Consumption quantity/(pcs)
December 2015	2
March 2016	3
February 2017	4
December 2018	2
January 2019	4
December 2020	3

Taking 90 ° galvanized threaded elbow/DN32 as an example, it can be seen from Table 3 that the six consumption records of the spare parts in the history are all winter, that is, the spare parts are defined as winter spare parts.

Seasonal materials have a strong cyclical attribute, and both procurement and consumption are based on a certain season. According to this feature, through the analysis of the historical data of seasonal materials, the standard normal verification of the demand is carried out according to the seasonal cycle (half a year), which conforms to the normal distribution.

D. Safety inventory model of key spare parts

The key spare parts in professional coal terminal A are characterized by low consumption frequency. Through the analysis of the data of the past 10 years, 80% of the key spare parts are less than once a year. There are too few data points, lacking the significance of mathematical statistics. On the other hand, most of the procurement of key spare parts are custom-made parts imported from abroad, which are highly variable due to various factors such as international trade relations and the operation of foreign manufacturers. Therefore, traditional mathematical models cannot adapt to such business scenarios. In this case of great uncertainty, the empirical value variable of minimum inventory of key spare parts is introduced as the main weight of safety stock calculation. The empirical value of the minimum inventory of key spare parts is the minimum inventory of spare parts required to ensure the normal operation of the equipment calculated by the on-site demand department based on the evaluation of the operation status of the on-site equipment by various technicians.

According to the above ideas, improve and optimize the safety inventory model of key spare parts:

$$SS = M * K + (1 - K) * z\sqrt{\sigma_d^2(\bar{L}) + \sigma_L^2(\bar{d})^2} \quad (2)$$

In the formula, M is minimum inventory empirical value. K is proportion coefficient. $\bar{L}$ Average value of lead time (month). $\bar{d}$ is monthly average demand. σ_d is the standard deviation of monthly demand d , which follows the same normal distribution as the average demand and is not related. σ_L is the standard deviation of the lead time (month) L , and the lead time obey the normal distribution. The number of standard deviations under a certain service level is the safety factor. And Z equals to 1.65.

The size of variable K directly affects the result of the safety stock value of critical spare parts. Therefore, it is necessary to conduct simulation verification on the K value. Select 200 key spare parts as samples and calculate the monthly average demand and standard deviation of the monthly average demand based on historical consumption records. Using the K value as the input parameter, calculate the safety stock values of each spare part for $K=0.6$, $K=0.7$, $K=0.8$, and $K=0.9$ in sequence. Compare the calculated safety stock with the demand and calculate the deviation amplitude. Table IV are some sample analysis data results from 200 key spare parts.

TABLE IV. SAFETY STOCK OF KEY SPARE PARTS UNDER DIFFERENT K VALUES IN THE SAMPLE

Material	K=0.6	K=0.7	K=0.8	K=0.9
Roller /(pcs)	1.86	2.32	3.13	3.53
Frequency converter/(pcs)	1.1	1.27	2.49	3.67
Encoder/(pcs)	1.42	1.8	4.32	4.5
Power cable winding motor/(pcs)	0.3	0.45	0.52	0.82

Traveling frequency converter/(pcs)	5.39	7.14	9.88	14.74
Vacuum circuit breaker/(pcs)	7.67	8.63	11.65	12.6
Bearing seat assembly/(pcs)	8.84	11.44	13.02	14.68
Coupling/(pcs)	2.7	3.17	4.1	7.73
Pitch reducer/(pcs)	0.11	0.17	0.25	0.28

By analyzing the deviation between the safety stock and demand of 200 critical spare parts, when $K=0.9$, the calculated deviation between the safety stock and demand is the smallest. Therefore, a suitable K value of 0.9 was determined.

E. Reorder level model

The determination of reorder level belongs to a method of inventory control, and the metrology of control lies in measuring the inventory quantity at a certain location and tracking the increase or decrease of inventory quantity. The tracking operation is completed through manual or computer-assisted methods. Inventory control has two modes: periodic inspection and continuity inspection. According to the framework agreement procurement and targeted procurement based procurement model of A professional coal terminal, we will adopt a phased inspection mode. The reorder level model under the phased inspection mode is as follows:

$$R = \bar{d} * (1 + P/2T) + SS \quad (3)$$

In the formula, $\bar{d}$ is Monthly average demand, P is Average inspection cycle, T is average operating cycle, SS is Safety Stock.

When constructing the safety inventory model earlier, the material consumption data has been statistically calculated and the corresponding safety inventory SS has been calculated for each item. At the A professional coal terminal, $P=T=30$. So the reordering level of the materials can be calculated. The following is the calculation result of reorder level for key spare parts in Table V.

TABLE V. SAMPLE KEY SPARE PARTS REORDER LEVEL

Supplies	Average demand	Safety stock	Reorder level
Roller/(pcs)	3.62	4.88	10.31
Frequency converter/(pcs)	2.98	5.13	9.60
Encoder/(pcs)	3.66	4.86	10.35
Power cable winding motor/(pcs)	0.74	1.12	2.23

Traveling frequency converter/(pcs)	12.06	15.6	33.69
Vacuum circuit breaker/(pcs)	15.64	13.44	36.90
Bearing seat assembly/(pcs)	16.18	17.5	41.77
Coupling/(pcs)	6.01	9.63	18.65
Pitch reducer/(pcs)	0.32	0.54	1.02

After calculating the safety stock SS and reorder level R , A professional coal terminal can adopt a mixed inventory control strategy (T , R , SS) in daily material inventory management. Check the inventory every T time, and if it is below the reorder point R , proceed with the purchase order. The order quantity is the safety stock quantity SS minus the existing stock quantity; If it is higher than the reorder point R , no order will be placed

IV. CONCLUSION

The research on the safety stock model in this paper adopts the research idea of focusing on specific industries and business scenarios, which is different from the traditional pure mathematical statistical analysis method, making it more targeted and practical. It is hoped that the research made in this paper will provide some guidance for port enterprises on how to reasonably set up safety stock.

REFERENCES

- [1] T. Aouam and K. Kumar, "On the effect of overtime and subcontracting on supply chain safety stocks," *Omega*, vol. 89, pp. 1-20, 2019.
- [2] M. S. Puga, S. Minner, and J. S. Tancr ez, "Two-stage supply chain design with safety stock placement decisions," *International Journal of Production Economics*, vol. 209, pp. 183-193, 2019.
- [3] S. C. Graves, and T. Schoenmeyr, "Strategic safety-stock placement in supply chains with capacity constraints," *Manufacturing & Service Operations Management*, vol. 18, pp. 445-460, 2016.
- [4] J. N. Gon alves, M. S. Carvalho, and P. Cortez, "Operations research models and methods for safety stock determination: A review," *Operations Research Perspectives*, vol. 7, pp. 100164, 2020.
- [5] K. Kumar, and T. Aouam, "Effect of setup time reduction on supply chain safety stocks," *Journal of manufacturing systems*, vol. 49, pp. 1-15, 2018.
- [6] D. Prak, R. Teunter, and A. Syntetos, "On the calculation of safety stocks when demand is forecasted," *European Journal of Operational Research*, vol. 256, pp. 454-461, 2017.
- [7] H. Lu, H. Wang, Y. Xie, and H. Li, "Construction material safety-stock determination under nonstationary stochastic demand and random supply yield," *IEEE transactions on engineering management*, vol. 63, pp. 201-212, 2016.
- [8] M. Braglia, D. Castellano, and M. Frosolini, "Safety stock management in single vendor-single buyer problem under VMI with consignment stock agreement," *International Journal of Production Economics*, vol. 154, pp. 16-31, 2014.
- [9] B. Urlu, and N. K. Erkip, "Safety stock placement for serial systems under supply process uncertainty," *Flexible Services and Manufacturing Journal*, vol. 32, pp. 395-424, 2020.
- [10] H. Gilani, and H. Sahebi, "Optimal Design and Operation of the green pistachio supply network: A robust possibilistic programming model," *Journal of Cleaner Production*, vol. 282, pp. 125212, 2021.